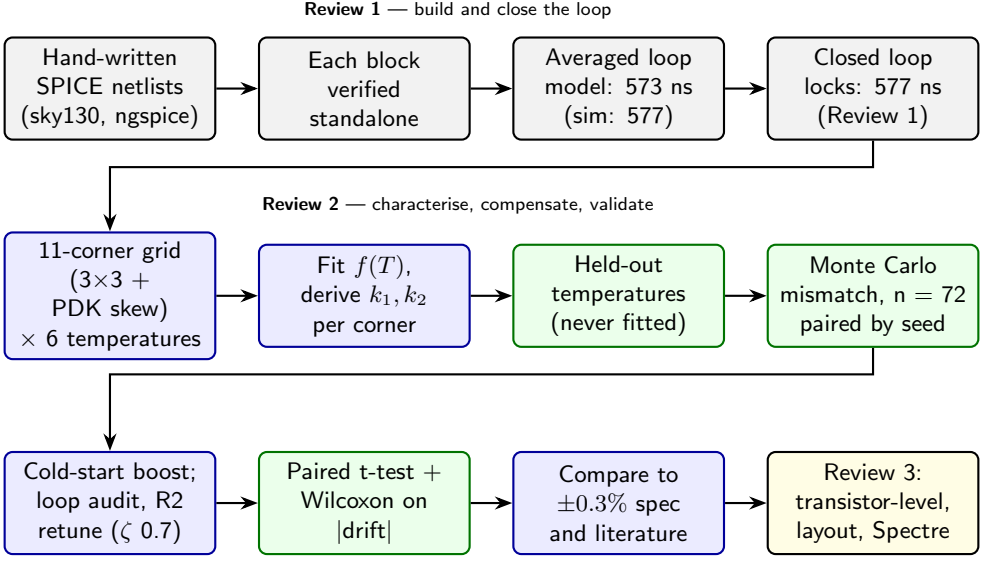

Review 1 — build and close the loop
Hand-written SPICE netlists (sky130, ngspice)
Each block verified standalone
Averaged loop model: 573 ns (sim: 577)
Closed loop locks: 577 ns (Review 1)
Review 2 — characterise, compensate, validate
11-corner grid (3×3 + PDK skew) × 6 temperatures
Fit $f(T)$, derive k_1, k_2 per corner
Held-out temperatures (never fitted)
Monte Carlo mismatch, n = 72 paired by seed
Cold-start boost; loop audit, R2 retune (ζ 0.7)
Paired t-test + Wilcoxon on |drift|
Compare to $\pm 0.3\%$ spec and literature
Review 3: transistor-level, layout, Spectre
grey = Review 1 blue = design step green = validation step (guards against fooling ourselves)